



SMILEI: A collaborative, open-source, multi-purpose particle-in-cell code for plasma simulation[☆]

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ARTICLE INFO

Article history:

Received 16 February 2017

Received in revised form 30 August 2017

Accepted 26 September 2017

Available online 10 October 2017

Keywords:

Plasma kinetic simulation

Particle-In-Cell (PIC)

High-performance computing

Laser-plasma interaction

Astrophysical plasmas

ABSTRACT

SMILEI is a collaborative, open-source, object-oriented (C++) particle-in-cell code. To benefit from the latest advances in high-performance computing (HPC), SMILEI is co-developed by both physicists and HPC experts. The code's structures, capabilities, parallelization strategy and performances are discussed. Additional modules (e.g. to treat ionization or collisions), benchmarks and physics highlights are also presented. Multi-purpose and evolutive, SMILEI is applied today to a wide range of physics studies, from relativistic laser-plasma interaction to astrophysical plasmas.

Program summary

Program title: SMILEI (version 3.2)

Program Files doi: <http://dx.doi.org/10.17632/gsn4x6mbrg.1>

Licensing provisions: This version of the code is distributed under the GNU General Public License v3

Programming language: C++11, Python 2.7

Nature of the problem: The kinetic simulation of plasmas is at the center of various physics studies, from laser-plasma interaction to astrophysics. To address today's challenges, a versatile simulation tool requires high-performance computing on massively parallel super-computers.

Solution method: The Vlasov–Maxwell system describing the self-consistent evolution of a collisionless plasma is solved using the Particle-In-Cell (PIC) method. Additional physics modules allow to account for additional effects such as collisions and/or ionization. A hybrid MPI-OpenMP strategy, based on a patch-based super-decomposition, allows for efficient cache-use, dynamic load balancing and high-performance on massively parallel super-computers.

Additional comments: Repository <https://github.com/SmileiPIC/Smilei>

References: <http://www.maisondelasimulation.fr/smilei>

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[☆] This paper and its associated computer program are available via the Computer Physics Communication homepage on ScienceDirect (<http://www.sciencedirect.com/science/journal/00104655>).

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1. Introduction

The Particle-In-Cell (PIC) approach was initially developed for fluid dynamics studies [1]. Having various advantages (conceptual simplicity, efficient implementation on massively parallel computers, etc.), it has become a central simulation tool for a wide range of physics studies, from semiconductors to cosmology or accelerator physics, and in particular to plasma physics. Today, the